The Quarter

An unrepresentable number, and why it is a choice

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The fourth card. It is the thing the other three compare.

“Can you find the polynomial?”

Reading note

This document is about one object. It cannot define that object by a formula, because a formula would be a representation and the object is the number that no record represents. What it can do is record, from cited results, that the object exists, that every record carries a projection of it, that no record computes it from its own contents, and that the only act which supplies it is a choice. Three projections are worked in full. Two bounds are then recorded, one from above and one from below, each of the form "a countable series, no finite truncation of which suffices." Then the argument is run the other way: if the quarter were comparable, the two famous questions become countable families of comparisons against it. Each statement carries one of three grades: *standard* (cited), *recorded*, or *proposed*. A *recorded* statement is a written sequence of steps whose every premise is a cited result or a definition of this card; the steps are a record, and that the record is mathematics, true and about its object, is the reader's reading of it, not something the record supplies. In that sense this card demonstrates no mathematics of its own: everything it rests on is cited, and everything it adds is record. The companion volumes are the geometric and numerical cards; the middle card is a construction checked by a proof assistant. Nothing from the middle card appears here by name; where its declarations are the evidence, that is said and graded *measured*. The three cards divide one argument: the geometric card fixes what must be retained, the numerical card fixes what may be claimed from a reading, and the middle card with the two record chapters checks that the record carries both. This card is about the one datum all three carry only as a projection with a receipt. The obstruction it describes is representational, not ontological: the root of the numerical card is an algebraic number with no decimal, the fourth coordinate of the geometric card is a real number with no image, and the name of the middle card is a choice with no derivation.

1 Records, projections, receipts

Definition 1.1 (Record)**.** A *record* is a finite sequence of symbols from a finite alphabet, together with a rule that extends it by exactly one symbol per step. The *length* of a record is the number of steps taken to write it.

Definition 1.2 (Representable)**.** A number is *representable* in a record system if some finite record determines it: a reader applied to that record returns the number and no other.

Definition 1.3 (Projection and receipt)**.** Let q be a datum and R a record. *R carries a projection of q* if R contains the image $\pi_R(q)$ of q under a map π_R that is not injective. A *receipt* for the projection is a further datum ρ such that $(\pi_R(q), \rho)$ determines q on a stated domain.

Definition 1.4 (The quarter)**.** A datum q is a *quarter* for a family of record systems if (i) every record in the family carries a projection of q; (ii) no record in the family determines q from its own contents, that is, q is not representable in any of them; (iii) a value of q is supplied, when it is supplied, by an act outside the record: a choice.

The definition says what the quarter is not. It does not say what it is, and Section 3 shows that this is forced.

2 Three projections

2.1 The integration constant

Fix the volume element of a four-dimensional Lorentzian metric by working in charts with $\sqrt{-g} = 1$, and vary the metric only along traceless directions. The stationary equations of the first-order vacuum action are then the traceless Einstein equations, and the contracted Bianchi identity forces the scalar curvature to be constant: $R = 4\Lambda$. The cosmological constant appears as a constant of integration, not as a coefficient of the action; this is the unimodular presentation of gravity, and the derivation is carried out in the numerical card beside its gravity model. *Standard.*

What the solution carries is $\Lambda g_{\mu\nu}$: the projection of the constant onto the field. The receipt is the datum that fixed it. Nothing in the equations produces Λ; the equations say only that once chosen it is carried unchanged. That is clause (i) and clause (ii) of Definition 1.4 for the field equations.

2.2 The omitted zero

Riemann's function is $\psi(x) = \sum_{n\geq 1} e^{-n^2\pi x}$ for $x > 0$ [12]. Jacobi's transformation of the theta function $\theta(x) = 1 + 2\psi(x) = \sum_{n\in\mathbb{Z}} e^{-n^2\pi x}$ is $\theta(1/x) = \sqrt{x}\,\theta(x)$, and in terms of ψ it reads

$$\psi(1/x) = \sqrt{x}\,\psi(x) + \frac{\sqrt{x}-1}{2}. \tag{1}$$

The term $(\sqrt{x}-1)/2$ is the $n = 0$ term that ψ omits and θ keeps. *Standard* [5, Ch. 1].

Proposition 2.1 (The omitted term is a residue)**.** *For* $\operatorname{Re} s > 1$,

$$\pi^{-s/2}\Gamma(s/2)\zeta(s) = \int_0^\infty \psi(x)\, x^{s/2-1}\,\mathrm{d}x,$$

and splitting the integral at $x = 1$ *and applying* (1) *on* $(0,1)$ *gives*

$$\pi^{-s/2}\Gamma(s/2)\zeta(s) = \frac{1}{s(s-1)} + \int_1^\infty \psi(x)\left(x^{s/2-1} + x^{(1-s)/2-1}\right)\mathrm{d}x.$$

The integral is entire and invariant under $s \mapsto 1-s$*. The poles of the left side at* $s = 0$ *and* $s = 1$*, and in particular the pole of* ζ *at* $s = 1$ *with residue* 1*, come entirely from the omitted term.* Standard; the computation is reproduced here.

Proof. Termwise, $\int_0^\infty e^{-n^2\pi x}x^{s/2-1}\,\mathrm{d}x = \pi^{-s/2}\Gamma(s/2)n^{-s}$, and the sum converges absolutely for $\operatorname{Re} s > 1$. On $(0,1)$ substitute $x \mapsto 1/x$ and use (1):

$$\int_0^1 \psi(x)x^{s/2-1}\,\mathrm{d}x = \int_1^\infty \psi(x)x^{(1-s)/2-1}\,\mathrm{d}x + \frac{1}{2}\int_1^\infty \left(x^{-s/2-1/2} - x^{-s/2-1}\right)\mathrm{d}x.$$

The last integral equals $\frac{1}{s-1} - \frac{1}{s} = \frac{1}{s(s-1)}$ for $\operatorname{Re} s > 1$. The remaining integrals converge for all s since $\psi(x) = O(e^{-\pi x})$, and their sum is symmetric under $s \mapsto 1-s$ by inspection. □

The projection carried by ζ is the residue 1 at $s = 1$, the image of the single term that was left out when the count began at one instead of zero. The receipt is the choice of where to start counting. The self-dual point $x = 1$ is where the two halves of the record read the same, and it is the only place the omitted term can be seen at all.

2.3 The simulation constant

Let U and V be universal machines and $K_U(x)$, $K_V(x)$ the lengths of the shortest programs producing x on each.

Proposition 2.2 (Invariance)**.** *There is a constant c_{UV}, depending on U and V and not on x, with $K_U(x) \leq K_V(x) + c_{UV}$ for all x.* Standard *[7, 9].*

Proof. c_{UV} is the length of a program for U that simulates V; prefix it to any V-program for x. □

Proposition 2.3 (No system certifies the constant)**.** *For every consistent formal system S with a computable set of axioms there is a constant c_S such that S proves no statement of the form $K_U(x) > c_S$, although such statements are true for all but finitely many x.* Standard *[2].*

Every record written on a universal machine carries a projection of c_{UV}: its lengths are correct up to that constant and no record on either machine fixes it. The constant is supplied by the choice of machine. Proposition 2.3 says that it is not merely unfixed but uncertifiable from inside, beyond a bound that depends on the system doing the certifying.

2.4 One object

Proposition 2.4 (The three projections differ by constants)**.** *Each of the three data above is carried by its record only up to an additive constant that the record does not fix: Λ up to the data that select the solution, the $n = 0$ term up to the origin of the count, c_{UV} up to the choice of machine. In each case the record is complete without the constant and incomplete with respect to it: the record's internal identities (the Bianchi identity, the functional equation, the invariance theorem) hold for every value, and single out none.* Recorded*, as a restatement of the three preceding cited results [5, 7, 9].*

That is what it means for them to be projections of one object rather than three objects. A projection is what survives a map that forgets exactly the constant; three records that each forget the same kind of thing are three views of the thing forgotten. *Proposed*: that the three constants are the same quarter. The claim is not that they are numerically equal, which would be meaningless across three units; it is that each is the image of a single act, the choice, under three readers.

3 The quarter is a choice

Theorem 3.1 (Unrepresentable means chosen)**.** *Let q satisfy clauses (i) and (ii) of Definition 1.4 for a family of record systems that is closed under finite extension. Then any procedure that supplies a value of q to a record in the family is not a procedure of that family: it is a choice, in the sense that it selects one element of a set the family can name but cannot enumerate to the point of selection.* Recorded*, from Definition 1.4 and the closure hypothesis; the step is the standard one, that a selection procedure inside the system would make the value representable there [2, 9].*

Proof. Suppose a procedure P expressible in the family supplied a value of q. Then the record consisting of P's description followed by its run is a finite record in the family, by closure under finite extension, and its reader returns q. That contradicts clause (ii). Hence no such P exists in

the family. The set of candidate values is nameable, since clause (i) says every record carries a projection and the fibre of the projection is a definite set. A selection from a nameable set with no expressible selection procedure is what a choice is. □

The theorem is the reason Definition 1.4 could only say what the quarter is not. A definition is a record. The middle card's proof assistant records the same fact in its own ledger: no step of its reading depends on the axiom of choice, and the one declaration that does reaches it only through an automatic arithmetic procedure, not through any selection the card makes. *Measured*, from a sweep of its declarations. The card keeps both extensionality principles, for propositions and for quotients, and refuses choice, the third ingredient of Diaconescu's theorem: choice together with the other two makes every proposition a yes-or-no question with an answer [4]. *Standard.* Every question the middle card answers, it answers by a decision it carries; it never borrows the answer.

4 Two bounds

Both bounds have the same form: a countable series of which every finite truncation falls short. The upper bound says every representable number is reached by a finite truncation. The lower bound says the demonstration of a minimal bound is not.

4.1 Upper bound: unique factorization

Theorem 4.1 (Fundamental theorem of arithmetic)**.** *Every integer $n \geq 2$ is a product of primes, uniquely up to order.* Standard *[6, Thm. 2].*

Proposition 4.2 (Representable numbers have short records)**.** *Write $\Omega(n)$ for the number of prime factors of n counted with multiplicity. Then $\Omega(n) \leq \log_2 n$, and the factorization record of n (its primes with multiplicities, each written in binary) has length at most $C \log_2 n \cdot \log_2 \log_2 n$ for an absolute constant C. Hence every representable integer is determined by a record whose length is a finite polynomial in the number of digits of n.* Recorded*, from the fundamental theorem of arithmetic [6, Thm. 2].*

Proof. Each prime factor is at least 2, so $2^{\Omega(n)} \leq n$. Each factor is at most n and so has at most $\log_2 n + 1$ binary digits; each multiplicity is at most $\log_2 n$. The record has at most $\Omega(n)$ entries of that size. □

Read as a generating series, the representable integers are the coefficients of finite polynomials: each n is reached by a finite product. The quarter is not among them. Its record, if it had one, would be the series itself, $\sum_k a_k x^k$ with no last term, and Theorem 3.1 says no finite truncation returns it. The upper bound is therefore exact for what it covers and silent on the quarter: *representable* is bounded above by a finite polynomial in the digits; *unrepresentable* is what the countable series carries and no polynomial does. *Recorded*, with the identification of the quarter as the series' limit *proposed*.

4.2 Lower bound: the cost of a minimal bound

The separation question asks whether every problem whose solutions can be checked in polynomial time can also be solved in polynomial time [3]. A proof of separation must exhibit, for some checkable problem, a lower bound on the size of every solver.

Proposition 4.3 (A certified minimal bound is uncertifiable)**.** *Let S be a consistent formal system with computable axioms and c_S its constant from Proposition 2.3. Any proof in S that proceeds by certifying, for infinitely many inputs x, that no description of a solver's behaviour on x is shorter than a stated bound exceeding c_S, does not exist.* Recorded*, from Proposition 2.3 [2].*

Proof. Such a certificate for x is a proof in S of $K_U(x') > c_S$ for the string x' encoding the solver's behaviour on x, which Proposition 2.3 excludes. □

This is the arithmetic content behind the barriers that the separation question is known to face [11]: a proof that works by pointing at a minimal representation and certifying its minimality is blocked at a constant the proving system cannot see past. The lower bound is therefore of the same shape as the upper one, with the inequality reversed. The demonstration of a minimal bound costs at least a countable series, no finite truncation of which certifies it; the number at which the series would close is c_S, and c_S is unrepresentable in S.

What this does not say must be said. It does not decide the separation question in either direction. A bound that cannot be certified is not a bound that fails; the constant that cannot be represented is not absent. It says that the question is not one a count settles, and that every record of an attempted answer carries the projection of a choice it did not make. *Recorded* for the negative statement; any reading of it as an answer to the question is *not proposed*, and is listed here so that it cannot be attributed to this document.

5 The other way

Sections 3 and 4 went one way: from the quarter's unrepresentability to the shape of two bounds. This section goes the other way. Suppose the quarter were *comparable*: suppose that, for each representable number, the question of whether it lies below or above the quarter were answered. Comparability is weaker than representability, since an oracle for the order need not produce the number, and it is exactly what the two famous questions ask for.

Theorem 5.1 (The Riemann hypothesis is a countable family of comparisons against an unrepresentable constant)**.** *Let $\sigma(n)$ be the sum of the divisors of n and γ Euler's constant. The Riemann hypothesis holds if and only if*

$$\sigma(n) < e^{\gamma}\, n \log\log n \qquad \text{for every } n > 5040.$$

Equivalently [8], with $H_n = \sum_{k \le n} 1/k$, if and only if $\sigma(n) \le H_n + e^{H_n} \log H_n$ for every $n \ge 1$, with equality only at $n = 1$. Standard *[13, 8].*

Theorem 5.2 (Positivity form)**.** *Let $\lambda_n = \sum_{\rho} \big[1 - (1 - 1/\rho)^n\big]$, the sum over the nontrivial zeros of ζ taken in conjugate pairs. The Riemann hypothesis holds if and only if $\lambda_n \ge 0$ for every $n \ge 1$.* Standard *[10].*

Each form is a countable sequence of comparisons, and in the first the thing compared against is e^{γ}, a constant with no finite record. The hypothesis says the quarter is comparable infinitely often and the answer never changes side. The self-dual line of Proposition 2.1 is where the omitted term is seen; the hypothesis says the entire part of the record has all its zeros there, which is the same symmetry read as an order. So, read through this document's definitions: the Riemann hypothesis is the assertion that the omitted-zero projection of the quarter is comparable, every time, to the same side. *Recorded*, as a reading of the two cited theorems; the hypothesis itself is *proposed*, as it is everywhere.

Theorem 5.3 (Relative to an oracle the separation question can close)**.** *There is an oracle A with* $\mathrm{P}^A = \mathrm{NP}^A$ *and an oracle B with* $\mathrm{P}^B \neq \mathrm{NP}^B$*.* Standard *[1].*

An oracle that answers comparisons against the quarter is an oracle. The lower bound of Proposition 4.3 said that, without one, the demonstration of a minimal bound is a countable series that never closes. Theorem 5.3 says that with one the series may close, and the collapse is then a theorem relative to that oracle. "Polynomial time of countable degree," time measured by a series rather than a polynomial, is time relative to the oracle that makes the quarter comparable: with every comparison answered, the countable series truncates, and relative to such an oracle P = NP is consistent. *Recorded*, from Theorem 5.3 and Proposition 4.3 [1, 2].

Proposition 5.4 (Both questions are statements about comparability)**.** *The Riemann hypothesis is equivalent to a sentence of the form "for every representable n, comparison(n) holds," with the comparison against an unrepresentable constant (Theorem 5.1). The statement* $\mathrm{P} = \mathrm{NP}$ *is equivalent to a sentence of the form "there is a representable procedure such that for every representable input the procedure succeeds in time bounded by a representable polynomial." The first has one universal quantifier over representables; the second has an existential quantifier in front of it. Supplying the quarter by a choice removes an existential quantifier: a chosen object is a given one.* Recorded*, as a statement of quantifier shape; the logical classification of the Riemann hypothesis as a universal arithmetic sentence is* standard *[8].*

Remark (The proposed equivalence)*.* The planning input for this card proposes that the two questions are one question read from two records: that the Riemann hypothesis holds if and only if the separation question closes in polynomial time of countable degree, each being the statement that the unrepresentable number is comparable. What is recorded above is weaker and should be kept distinct from it. Both questions are countable families of comparisons against the quarter. The first asks whether the comparisons all fall one way; the second asks whether a chosen comparison oracle closes a series. The choice that would supply the quarter is the same act in both, and it is the act neither record performs. Whether the two families of comparisons are the same family under some reader is *proposed* and not derived here. What would discharge it is a reader taking each comparison of Theorem 5.1 to a step of a bounded trace and back, with the oracle of Theorem 5.3 as the quarter in both. Until then the honest sentence is the one both records already agree on: neither question is settled by a count, and each is settled, if at all, by a comparison with the number the count cannot write.

The last question

Every question the middle card answers comes with a receipt that a checker reads in time proportional to its length: a search writes the ledger down, and a check confirms it line by line. That is a

certificate and a polynomial verifier. *Measured*, from the card's construction. The question is what lies past every such receipt.

Proposition 5.5 (A new question has no polynomial verifier)**.** *Suppose a record system answers satisfiability and is closed under polynomial-time reduction. Then any yes-or-no question it cannot already answer is either decidable and outside NP, or undecidable. In either case it has no polynomial verifier.* Recorded*, from the Cook–Levin theorem [3] and the definition of NP.*

Proof. If the question were in NP it would reduce in polynomial time to satisfiability, which the system answers, so the system would answer it. Hence it is not in NP, which by definition means that its yes-instances have no certificate of polynomial size checkable in polynomial time. An undecidable question is outside NP a fortiori, since every question in NP is decided by trying every certificate of the bounded size; so again there is no polynomial verifier. □

Three things this does not say must be said. First, new questions can still be composed: negation alone turns satisfiability into its complement, which is a question in co-NP. It falls under the proposition as a question outside NP unless NP and co-NP coincide, and that is open; showing any particular question has no polynomial verifier would separate them, and with them P from NP. *Standard.* Second, the premise is not established for the middle card. Its correspondence between traces and formulas is proved for every trace, but for one step relation, exclusive or within a cell; realizing satisfiability in general would require it to encode the step of an arbitrary verifier. *Recorded*, as owed. Third, the proposition is not demonstrated by the middle card's proof assistant, and this card does not expect it to be: it is a statement about every question and every verifier, quantified over the machines that would check it, and the checker cannot certify its own boundary (Proposition 2.3). It is recorded here from cited results, and the reader reads it.

Read through this document's definitions: a question with no polynomial verifier is one no record can carry a receipt for. If its answer is supplied, it is supplied as the quarter is, by a choice the record does not make. *Proposed.*

6 Where the quarter is in the three cards

The geometric card carries the whole in four coordinates and reads three. The fourth survives as the shadow inside the ball and a sign that says which sheet; drop the sign and two states have one image. Its quarter is the fourth coordinate; its receipt is the sign. The numerical card carries a certified reading to twenty-four places and a cosmological constant as a residue. Its quarter is the root itself, $972(6+\sqrt{10})/65$: finitely representable as an algebraic number, never as a decimal, so its digits are a countable series that does not close. The seed chooses which root; the two rational walls of the bracket are the receipt; the twenty-four printed places are the projection, certified exactly to the resolution at which the number stops being representable and no further. The datum that fixed Λ is its second quarter, with the Bianchi identity as receipt. *Recorded* for the bracket and the truncation, from the numerical card's certificate; the reading of the root as the quarter is *proposed.* The middle card carries a tape whose every cell is the same symbol, and a census that reads only what has a name. Its quarter is the name: the compiler can require one and cannot mint one. In all three the record is complete without the quarter and incomplete with respect to it, which is Proposition 2.4 once more.

This card has no press check, because a press check compares two readings of a register and the quarter is what the readings are of. It has no counter, because nothing is written here that was not already carried. It has no page one.

The last line. The number the three cards straddle resembles a measured constant to four places. As far as we can tell, there is no mathematical representation of the derivation of the fine structure constant. That sentence is graded *standard*, as a report on the literature: the constant is measured, and no derivation of it is accepted. The resemblance is graded *proposed*, as it is throughout. What this card adds is only the observation that a number with no derivation and a number with no finite record are the same kind of thing to a record: a projection, a receipt, and a choice.

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